All Course Materials

BC II / III Syllabus (https://imsa.instructure.com/courses/2632/files/858779?wrap=1) (https://imsa.instructure.com/courses/2632/files/858779/download?download_frd=1)

BC II / III Learning Targets and Grade Record (https://imsa.instructure.com/courses/2632/files/858778?wrap=1) (https://imsa.instructure.com/courses/2632/files/858778/download?download_frd=1)

Textbook

Calculus, Volume 1 (https://openstax.org/details/books/calculus-volume-1) (BC I/II Material)

Calculus, Volume 2 (https://openstax.org/details/books/calculus-volume-2) (BC II/III Material)

Online Resources

https://www.derivative-calculator.net/ (https://www.derivative-calculator.net/) (Calculates derivatives, showing steps)

https://www.integral-calculator.com/ (https://www.integral-calculator.com/) (Calculates integrals, showing steps)

Paul's Online Calculus Notes (https://tutorial.math.lamar.edu/classes/calci/calci.aspx) (Includes many worked-out examples and additional exercises with solutions)

Handouts

Unit 1 - Completeness

- **1.0 The Need for Precise Definitions and Proof.pdf (https://imsa.instructure.com/courses/2632/files/858465?wrap=1) (https://imsa.instructure.com/courses/2632/files/858465/download?download_frd=1)**
- **1.1 Real Numbers.pdf (https://imsa.instructure.com/courses/2632/files/858131?wrap=1) (https://imsa.instructure.com/courses/2632/files/858131/download?download_frd=1)**
- **1.2 Rational and Irrational Numbers.pdf (https://imsa.instructure.com/courses/2632/files/858323?wrap=1) (https://imsa.instructure.com/courses/2632/files/858323/download?download_frd=1)**
- **1.3 Completeness.pdf (https://imsa.instructure.com/courses/2632/files/858446?wrap=1) (https://imsa.instructure.com/courses/2632/files/858446/download?download_frd=1)**
- **1.4 Consequences of Completeness.pdf (https://imsa.instructure.com/courses/2632/files/858445?wrap=1) (https://imsa.instructure.com/courses/2632/files/858445/download?download_frd=1)**

- 1.5 Nested Intervals Theorem.pdf (https://imsa.instructure.com/courses/2632/files/858448?wrap=1) (https://imsa.instructure.com/courses/2632/files/858448/download?download_frd=1)
- 1.6 Two Important Theorems.pdf (https://imsa.instructure.com/courses/2632/files/858447?wrap=1) (https://imsa.instructure.com/courses/2632/files/858447/download?download_frd=1)
- 1.7 The Bisection Method.pdf (https://imsa.instructure.com/courses/2632/files/858453?wrap=1) (https://imsa.instructure.com/courses/2632/files/858453/download?download_frd=1)

Unit 2 - Limits

- 2.1 A Preview of Calculus.pdf (https://imsa.instructure.com/courses/2632/files/858335?wrap=1) (https://imsa.instructure.com/courses/2632/files/858335/download?download_frd=1)
- 2.2 Intro to Limits.pdf (https://imsa.instructure.com/courses/2632/files/858037?wrap=1) (https://imsa.instructure.com/courses/2632/files/858037/download?download_frd=1)
- 2.3 Estimating Limits.pdf (https://imsa.instructure.com/courses/2632/files/858473?wrap=1) (https://imsa.instructure.com/courses/2632/files/858473/download?download_frd=1)
- 2.4 Precise Definition of a Limit.pdf (https://imsa.instructure.com/courses/2632/files/858307?wrap=1) (https://imsa.instructure.com/courses/2632/files/858307/download?download_frd=1)
- 2.5 Calculating Limits Analytically.pdf (https://imsa.instructure.com/courses/2632/files/858025?wrap=1) (https://imsa.instructure.com/courses/2632/files/858025/download?download_frd=1)
- 2.6 The Limit Laws.pdf (https://imsa.instructure.com/courses/2632/files/858362?wrap=1) (https://imsa.instructure.com/courses/2632/files/858362/download?download_frd=1)
- 2.7 Limit Laws Practice.pdf (https://imsa.instructure.com/courses/2632/files/858357?wrap=1) (https://imsa.instructure.com/courses/2632/files/858357/download?download_frd=1)
- 2.8 Additional Limit Evaluation Techniques.pdf (https://imsa.instructure.com/courses/2632/files/858472?wrap=1) (https://imsa.instructure.com/courses/2632/files/858472/download?download_frd=1)
- 2.9 More Limits.pdf (https://imsa.instructure.com/courses/2632/files/858363?wrap=1) (https://imsa.instructure.com/courses/2632/files/858363/download?download_frd=1)
- 2.10 One-sided, infinite limits, limits at infinity calculations.pdf (https://imsa.instructure.com/courses/2632/files/858287?wrap=1) (https://imsa.instructure.com/courses/2632/files/858287/download?download_frd=1)
- 2.11 Limits of trig and exponential functions.pdf (https://imsa.instructure.com/courses/2632/files/858294?wrap=1) (https://imsa.instructure.com/courses/2632/files/858294/download?download_frd=1)
- 2.12 Squeeze Theorem.pdf (https://imsa.instructure.com/courses/2632/files/858181?wrap=1) (https://imsa.instructure.com/courses/2632/files/858181/download?download_frd=1)

Unit 3 - Continuity

- 3.1 Continuity.pdf (https://imsa.instructure.com/courses/2632/files/858364?wrap=1) (https://

imsa.instructure.com/courses/2632/files/858364/download?download_frd=1)

- 3.2 Limits of inverse functions and compositions.pdf (https://imsa.instructure.com/courses/2632/files/858474?wrap=1) (https://imsa.instructure.com/courses/2632/files/858474/download?download_frd=1)
- 3.3 Intermediate Value Theorem.pdf (https://imsa.instructure.com/courses/2632/files/858365?wrap=1) (https://imsa.instructure.com/courses/2632/files/858365/download?download_frd=1)
- 3.4 Proof of the Intermediate Value Theorem.pdf (https://imsa.instructure.com/courses/2632/files/858277?wrap=1) (https://imsa.instructure.com/courses/2632/files/858277/download?download_frd=1)

Unit 4 - Derivatives

- 4.1 Introduction to Derivatives.pdf (https://imsa.instructure.com/courses/2632/files/858262?wrap=1) (https://imsa.instructure.com/courses/2632/files/858262/download?download_frd=1)
- 4.2 Definition of the Derivative.pdf (https://imsa.instructure.com/courses/2632/files/858369?wrap=1) (https://imsa.instructure.com/courses/2632/files/858369/download?download_frd=1)
- 4.3 Definition of Derivatives Practice.pdf (https://imsa.instructure.com/courses/2632/files/858373?wrap=1) (https://imsa.instructure.com/courses/2632/files/858373/download?download_frd=1)
- 4.4 The Derivative as a Function.pdf (https://imsa.instructure.com/courses/2632/files/858272?wrap=1) (https://imsa.instructure.com/courses/2632/files/858272/download?download_frd=1)
- 4.5 Power, Sum, Constant Multiple Rules.pdf (https://imsa.instructure.com/courses/2632/files/858023?wrap=1) (https://imsa.instructure.com/courses/2632/files/858023/download?download_frd=1)
- 4.6 Derivatives of Sine, Cosine, Exp.pdf (https://imsa.instructure.com/courses/2632/files/858372?wrap=1) (https://imsa.instructure.com/courses/2632/files/858372/download?download_frd=1)
- 4.7 The Product Rule.pdf (https://imsa.instructure.com/courses/2632/files/858441?wrap=1) (https://imsa.instructure.com/courses/2632/files/858441/download?download_frd=1)
- 4.8 The Quotient Rule.pdf (https://imsa.instructure.com/courses/2632/files/858435?wrap=1) (https://imsa.instructure.com/courses/2632/files/858435/download?download_frd=1)
- 4.9 The Chain Rule.pdf (https://imsa.instructure.com/courses/2632/files/858450?wrap=1) (https://imsa.instructure.com/courses/2632/files/858450/download?download_frd=1)
- 4.10 Proof of the Chain Rule.pdf (https://imsa.instructure.com/courses/2632/files/858452?wrap=1) (https://imsa.instructure.com/courses/2632/files/858452/download?download_frd=1)
- 4.11 Related Rates.pdf (https://imsa.instructure.com/courses/2632/files/858132?wrap=1) (https://imsa.instructure.com/courses/2632/files/858132/download?download_frd=1)
- 4.12 Related Rates Practice.pdf (https://imsa.instructure.com/courses/2632/files/858454?wrap=1) (https://imsa.instructure.com/courses/2632/files/858454/download?download_frd=1) (Related Rates Solution.pdf (https://imsa.instructure.com/courses/2632/files/858475?wrap=1)

(https://imsa.instructure.com/courses/2632/files/858475/download?download_frd=1))

- 4.13 Implicit Differentiation.pdf (https://imsa.instructure.com/courses/2632/files/858482?wrap=1) (https://imsa.instructure.com/courses/2632/files/858482/download?download_frd=1)
- 4.14 Implicit Differentiation Practice.pdf (https://imsa.instructure.com/courses/2632/files/858285?wrap=1) (https://imsa.instructure.com/courses/2632/files/858285/download?download_frd=1)
- 4.15 Derivatives of Exponential and Logarithmic Functions.pdf (https://imsa.instructure.com/courses/2632/files/858374?wrap=1) (https://imsa.instructure.com/courses/2632/files/858374/download?download_frd=1)
- 4.16 Logarithmic Differentiation.pdf (https://imsa.instructure.com/courses/2632/files/858301?wrap=1) (https://imsa.instructure.com/courses/2632/files/858301/download?download_frd=1)
- 4.17 Derivatives of Inverse Functions.pdf (https://imsa.instructure.com/courses/2632/files/858302?wrap=1) (https://imsa.instructure.com/courses/2632/files/858302/download?download_frd=1)
- 4.18 Derivatives of Inverse Trig Functions.pdf (https://imsa.instructure.com/courses/2632/files/858451?wrap=1) (https://imsa.instructure.com/courses/2632/files/858451/download?download_frd=1)

Unit 5 - Applications of Derivatives

- 5.1 Linear Approximation and Differentials.pdf (https://imsa.instructure.com/courses/2632/files/858483?wrap=1) (https://imsa.instructure.com/courses/2632/files/858483/download?download_frd=1)
  - Supplement to 5.1 Linear Approximation and Differentials.pdf (https://imsa.instructure.com/courses/2632/files/858488?wrap=1) (https://imsa.instructure.com/courses/2632/files/858488/download?download_frd=1)
- 5.2 Propagation of Error.pdf (https://imsa.instructure.com/courses/2632/files/858476?wrap=1) (https://imsa.instructure.com/courses/2632/files/858476/download?download_frd=1)
- 5.3 Maximum and Minimum Values.pdf (https://imsa.instructure.com/courses/2632/files/858480?wrap=1) (https://imsa.instructure.com/courses/2632/files/858480/download?download_frd=1)
- 5.4 Closed Interval Method Practice.pdf (https://imsa.instructure.com/courses/2632/files/858467?wrap=1) (https://imsa.instructure.com/courses/2632/files/858467/download?download_frd=1)
- 5.5 Proof of the EVT.pdf (https://imsa.instructure.com/courses/2632/files/858468?wrap=1) (https://imsa.instructure.com/courses/2632/files/858468/download?download_frd=1)
- 5.6 MVT Intro.pdf (https://imsa.instructure.com/courses/2632/files/858491?wrap=1) (https://imsa.instructure.com/courses/2632/files/858491/download?download_frd=1)
- 5.7 Mean Value Theorem.pdf (https://imsa.instructure.com/courses/2632/files/858489?wrap=1) (https://imsa.instructure.com/courses/2632/files/858489/download?download_frd=1)

- 5.8 Mean Value Theorem Problem Set.pdf (https://imsa.instructure.com/courses/2632/files/858370?wrap=1) (https://imsa.instructure.com/courses/2632/files/858370/download?download_frd=1)
- 5.9 Corollaries of the MVT.pdf (https://imsa.instructure.com/courses/2632/files/858492?wrap=1) (https://imsa.instructure.com/courses/2632/files/858492/download?download_frd=1)
- 5.10 Intervals of Inc / Dec, First Derivative Test.pdf (https://imsa.instructure.com/courses/2632/files/858299?wrap=1) (https://imsa.instructure.com/courses/2632/files/858299/download?download_frd=1)
- 5.11 Concavity and Inflection Points.pdf (https://imsa.instructure.com/courses/2632/files/858490?wrap=1) (https://imsa.instructure.com/courses/2632/files/858490/download?download_frd=1)
- 5.12 Graphing Functions.pdf (https://imsa.instructure.com/courses/2632/files/858376?wrap=1) (https://imsa.instructure.com/courses/2632/files/858376/download?download_frd=1)
- 5.13 Graphing Practice.pdf (https://imsa.instructure.com/courses/2632/files/858306?wrap=1) (https://imsa.instructure.com/courses/2632/files/858306/download?download_frd=1)
- 5.14 Relationships among f,f',f''.pdf (https://imsa.instructure.com/courses/2632/files/858439?wrap=1) (https://imsa.instructure.com/courses/2632/files/858439/download?download_frd=1)
- 5.15 Optimization I.pdf (https://imsa.instructure.com/courses/2632/files/858359?wrap=1) (https://imsa.instructure.com/courses/2632/files/858359/download?download_frd=1) (Answers: See the textbook)
- 5.16 Optimization II.pdf (https://imsa.instructure.com/courses/2632/files/858377?wrap=1) (https://imsa.instructure.com/courses/2632/files/858377/download?download_frd=1) (Answers to Optimization II.pdf (https://imsa.instructure.com/courses/2632/files/858500?wrap=1) (https://imsa.instructure.com/courses/2632/files/858500/download?download_frd=1))
- 5.17 Optimization III.pdf (https://imsa.instructure.com/courses/2632/files/858378?wrap=1) (https://imsa.instructure.com/courses/2632/files/858378/download?download_frd=1) (Optimization III Answers.pdf (https://imsa.instructure.com/courses/2632/files/858501?wrap=1) (https://imsa.instructure.com/courses/2632/files/858501/download?download_frd=1))
- 5.18 Optimization Practice.pdf (https://imsa.instructure.com/courses/2632/files/858502?wrap=1) (https://imsa.instructure.com/courses/2632/files/858502/download?download_frd=1)
- 5.19 L'Hopital's Rule.pdf (https://imsa.instructure.com/courses/2632/files/858375?wrap=1) (https://imsa.instructure.com/courses/2632/files/858375/download?download_frd=1) (5.19 L'Hopital's Rule (Solution).pdf (https://imsa.instructure.com/courses/2632/files/858504?wrap=1) (https://imsa.instructure.com/courses/2632/files/858504/download?download_frd=1))
- 5.20 L'Hopital's Practice.pdf (https://imsa.instructure.com/courses/2632/files/858503?wrap=1) (https://imsa.instructure.com/courses/2632/files/858503/download?download_frd=1)
- 5.21 Proof of L'Hopital's Rule.pdf (https://imsa.instructure.com/courses/2632/files/858303?wrap=1) (https://imsa.instructure.com/courses/2632/files/858303/download?download_frd=1) (5.21 Proof of L'Hopital's Rule (Solution).pdf (https://imsa.instructure.com/courses/2632/

files/858505?wrap=1) (https://imsa.instructure.com/courses/2632/files/858505/download?download_frd=1))

Unit 6 - Integration

- 6.1 Distance from velocity.pdf (https://imsa.instructure.com/courses/2632/files/858507?wrap=1) (https://imsa.instructure.com/courses/2632/files/858507/download?download_frd=1)
- 6.2 The definite integral.pdf (https://imsa.instructure.com/courses/2632/files/858506?wrap=1) (https://imsa.instructure.com/courses/2632/files/858506/download?download_frd=1)
- 6.3 The Fundamental Theorem of Calculus.pdf (https://imsa.instructure.com/courses/2632/files/858508?wrap=1) (https://imsa.instructure.com/courses/2632/files/858508/download?download_frd=1)
- 6.4 Construction of the Integral.pdf (See POTW #11 (https://imsa.instructure.com/courses/2632/files/858509?wrap=1) (https://imsa.instructure.com/courses/2632/files/858509/download?download_frd=1))
- 6.5 Verification of the axioms.pdf (See POTW #12 (https://imsa.instructure.com/courses/2632/files/858510?wrap=1) (https://imsa.instructure.com/courses/2632/files/858510/download?download_frd=1))
- 6.6 Integration Formulas.pdf (https://imsa.instructure.com/courses/2632/files/858382?wrap=1) (https://imsa.instructure.com/courses/2632/files/858382/download?download_frd=1)
- 6.7 Substitution.pdf (https://imsa.instructure.com/courses/2632/files/858515?wrap=1) (https://imsa.instructure.com/courses/2632/files/858515/download?download_frd=1)
- 6.8 Integrals of Exponential Functions and 1:x.pdf (https://imsa.instructure.com/courses/2632/files/858514?wrap=1) (https://imsa.instructure.com/courses/2632/files/858514/download?download_frd=1)
- 6.9 Integral of Trig Functions.pdf (https://imsa.instructure.com/courses/2632/files/858440?wrap=1) (https://imsa.instructure.com/courses/2632/files/858440/download?download_frd=1)
- 6.10 More Trig Integrals.pdf (https://imsa.instructure.com/courses/2632/files/858305?wrap=1) (https://imsa.instructure.com/courses/2632/files/858305/download?download_frd=1)
- 6.11 Integration by Parts.pdf (https://imsa.instructure.com/courses/2632/files/858298?wrap=1) (https://imsa.instructure.com/courses/2632/files/858298/download?download_frd=1)
- 6.12 Integration by Parts Practice.pdf (https://imsa.instructure.com/courses/2632/files/858495?wrap=1) (https://imsa.instructure.com/courses/2632/files/858495/download?download_frd=1)
- 6.13 Even More Trig Integrals.pdf (https://imsa.instructure.com/courses/2632/files/858304?wrap=1) (https://imsa.instructure.com/courses/2632/files/858304/download?download_frd=1)
- 6.14 Trig Substitution.pdf (https://imsa.instructure.com/courses/2632/files/858496?wrap=1) (https://imsa.instructure.com/courses/2632/files/858496/download?download_frd=1)
- 6.15 Advanced Trig Substitution.pdf (https://imsa.instructure.com/courses/2632/files/858497?wrap=1) (https://imsa.instructure.com/courses/2632/files/858497/download?download_frd=1)

- 6.16 Integration of Rational Functions.pdf (https://imsa.instructure.com/courses/2632/files/858387?wrap=1) (https://imsa.instructure.com/courses/2632/files/858387/download?download_frd=1)
- 6.17 Variations of Integration by Partial Fractions.pdf (https://imsa.instructure.com/courses/2632/files/858384?wrap=1) (https://imsa.instructure.com/courses/2632/files/858384/download?download_frd=1)

Unit 7 - Approximate Integration

- 7.1 Riemann Sums.pdf (https://imsa.instructure.com/courses/2632/files/886110?wrap=1) (https://imsa.instructure.com/courses/2632/files/886110/download?download_frd=1)
- 7.2 Calculating Riemann Sums.pdf (https://imsa.instructure.com/courses/2632/files/858312?wrap=1) (https://imsa.instructure.com/courses/2632/files/858312/download?download_frd=1)
- 7.3 Working with Sums.pdf (https://imsa.instructure.com/courses/2632/files/858316?wrap=1) (https://imsa.instructure.com/courses/2632/files/858316/download?download_frd=1)
- 7.4 Definite Integrals as Limits of Riemann Sums.pdf (https://imsa.instructure.com/courses/2632/files/858388?wrap=1) (https://imsa.instructure.com/courses/2632/files/858388/download?download_frd=1)
- 7.5 Approximate Integration - Rectangles.pdf (https://imsa.instructure.com/courses/2632/files/858391?wrap=1) (https://imsa.instructure.com/courses/2632/files/858391/download?download_frd=1)
- 7.6 Approximate Integration - Trapezoids.pdf (https://imsa.instructure.com/courses/2632/files/858320?wrap=1) (https://imsa.instructure.com/courses/2632/files/858320/download?download_frd=1)
- 7.7 Approximate Integration - Parabolas.pdf (https://imsa.instructure.com/courses/2632/files/858321?wrap=1) (https://imsa.instructure.com/courses/2632/files/858321/download?download_frd=1)
- Proofs of Error Bound Formulas
  - 7.8a Proof of Left-endpoint Error Bound.pdf (https://imsa.instructure.com/courses/2632/files/858029?wrap=1) (https://imsa.instructure.com/courses/2632/files/858029/download?download_frd=1)
  - 7.8b Proof of Right-endpoint Error Bound.pdf (https://imsa.instructure.com/courses/2632/files/858389?wrap=1) (https://imsa.instructure.com/courses/2632/files/858389/download?download_frd=1)
  - 7.8c Proof of Error Bound for Midpoint Rule.pdf (https://imsa.instructure.com/courses/2632/files/858390?wrap=1) (https://imsa.instructure.com/courses/2632/files/858390/download?download_frd=1)
  - 7.8d Proof of Error Bound for Trapezoidal Approximation.pdf (https://imsa.instructure.com/courses/2632/files/858394?wrap=1) (https://imsa.instructure.com/courses/2632/files/858394/download?download_frd=1)

- 7.8e Proof of Error Bound for Simpson's Rule.pdf (https://imsa.instructure.com/courses/2632/files/858024?wrap=1) (https://imsa.instructure.com/courses/2632/files/858024/download?download_frd=1)

Unit 8 - Applications of Integration

- 8.1 Area Between Curves.pdf (https://imsa.instructure.com/courses/2632/files/858396?wrap=1) (https://imsa.instructure.com/courses/2632/files/858396/download?download_frd=1)
- 8.2 Volumes - The Disk Method (https://imsa.instructure.com/courses/2632/files/858385?wrap=1) (https://imsa.instructure.com/courses/2632/files/858385/download?download_frd=1)
- 8.3 Volumes - The Washer Method.pdf (https://imsa.instructure.com/courses/2632/files/858386?wrap=1) (https://imsa.instructure.com/courses/2632/files/858386/download?download_frd=1)
- 8.4 Volumes - The Method of Cylindrical Shells.pdf (https://imsa.instructure.com/courses/2632/files/858318?wrap=1) (https://imsa.instructure.com/courses/2632/files/858318/download?download_frd=1)
- 8.5 Volumes - Other Solids.pdf (https://imsa.instructure.com/courses/2632/files/858397?wrap=1) (https://imsa.instructure.com/courses/2632/files/858397/download?download_frd=1)
- 8.6 Arc Length of a Curve and Surface Area.pdf (https://imsa.instructure.com/courses/2632/files/858392?wrap=1) (https://imsa.instructure.com/courses/2632/files/858392/download?download_frd=1)
- 8.7 Mass and Density.pdf (https://imsa.instructure.com/courses/2632/files/858398?wrap=1) (https://imsa.instructure.com/courses/2632/files/858398/download?download_frd=1)
- 8.8 Work.pdf (https://imsa.instructure.com/courses/2632/files/858368?wrap=1) (https://imsa.instructure.com/courses/2632/files/858368/download?download_frd=1)
- 8.9 Average Value of a Function.pdf (https://imsa.instructure.com/courses/2632/files/858399?wrap=1) (https://imsa.instructure.com/courses/2632/files/858399/download?download_frd=1)

Unit 9 - Improper Integrals

- 9.1 Improper Integrals of Type 1.pdf (https://imsa.instructure.com/courses/2632/files/858404?wrap=1) (https://imsa.instructure.com/courses/2632/files/858404/download?download_frd=1)
- 9.2 Improper Integrals of Type 2.pdf (https://imsa.instructure.com/courses/2632/files/858400?wrap=1) (https://imsa.instructure.com/courses/2632/files/858400/download?download_frd=1)
- 9.3 Comparison Test for Improper Integrals.pdf (https://imsa.instructure.com/courses/2632/files/858401?wrap=1) (https://imsa.instructure.com/courses/2632/files/858401/download?download_frd=1)
- 9.4 Limit Comparison Test for Improper Integrals.pdf (https://imsa.instructure.com/courses/2632/files/858393?wrap=1) (https://imsa.instructure.com/courses/2632/files/858393/download?download_frd=1)

Unit 10 - Differential Equations

- 10.1 Differential Equations.pdf (https://imsa.instructure.com/courses/2632/files/858402?wrap=1) (https://imsa.instructure.com/courses/2632/files/858402/download?download_frd=1)
- 10.2 Direction Fields.pdf (https://imsa.instructure.com/courses/2632/files/858403?wrap=1) (https://imsa.instructure.com/courses/2632/files/858403/download?download_frd=1)
- 10.3 Euler's Method.pdf (https://imsa.instructure.com/courses/2632/files/858405?wrap=1) (https://imsa.instructure.com/courses/2632/files/858405/download?download_frd=1)
- 10.4 Separable Equations.pdf (https://imsa.instructure.com/courses/2632/files/858407?wrap=1) (https://imsa.instructure.com/courses/2632/files/858407/download?download_frd=1)
- 10.5 Orthogonal Trajectories.pdf (https://imsa.instructure.com/courses/2632/files/903008?wrap=1) (https://imsa.instructure.com/courses/2632/files/903008/download?download_frd=1)
- 10.6 Mixing Problems.pdf (https://imsa.instructure.com/courses/2632/files/858411?wrap=1) (https://imsa.instructure.com/courses/2632/files/858411/download?download_frd=1)
- 10.7 Population Growth.pdf (https://imsa.instructure.com/courses/2632/files/858310?wrap=1) (https://imsa.instructure.com/courses/2632/files/858310/download?download_frd=1)
- 10.8 Logistic Growth.pdf (https://imsa.instructure.com/courses/2632/files/858413?wrap=1) (https://imsa.instructure.com/courses/2632/files/858413/download?download_frd=1)
- 10.9 Logistic Practice.pdf (https://imsa.instructure.com/courses/2632/files/858395?wrap=1) (https://imsa.instructure.com/courses/2632/files/858395/download?download_frd=1)

Unit 11 - Sequences and Series

- 11.1 Sequences.pdf (https://imsa.instructure.com/courses/2632/files/858409?wrap=1) (https://imsa.instructure.com/courses/2632/files/858409/download?download_frd=1)
- 11.2 Bounded Monotonic Sequence Theorem.pdf (https://imsa.instructure.com/courses/2632/files/858406?wrap=1) (https://imsa.instructure.com/courses/2632/files/858406/download?download_frd=1)
- 11.3 Series.pdf (https://imsa.instructure.com/courses/2632/files/858326?wrap=1) (https://imsa.instructure.com/courses/2632/files/858326/download?download_frd=1)
- 11.4 The Divergence Test.pdf (https://imsa.instructure.com/courses/2632/files/858327?wrap=1) (https://imsa.instructure.com/courses/2632/files/858327/download?download_frd=1)
- 11.5 The Integral Test.pdf (https://imsa.instructure.com/courses/2632/files/858328?wrap=1) (https://imsa.instructure.com/courses/2632/files/858328/download?download_frd=1)
- 11.6 Estimating the Sum of a Series.pdf (https://imsa.instructure.com/courses/2632/files/858410?wrap=1) (https://imsa.instructure.com/courses/2632/files/858410/download?download_frd=1)
- 11.7 The Comparison Tests.pdf (https://imsa.instructure.com/courses/2632/files/858332?wrap=1) (https://imsa.instructure.com/courses/2632/files/858332/download?download_frd=1)
- 11.8 Using the Comparison Test to Estimate Sums.pdf (https://imsa.instructure.com/courses/2632/files/858333?wrap=1) (https://imsa.instructure.com/courses/2632/files/858333/download?download_frd=1)

- 11.9 Alternating Series.pdf (https://imsa.instructure.com/courses/2632/files/858334?wrap=1) (https://imsa.instructure.com/courses/2632/files/858334/download?download_frd=1)
- 11.10 Absolute Convergence and the Ratio and Root Tests.pdf (https://imsa.instructure.com/courses/2632/files/858338?wrap=1) (https://imsa.instructure.com/courses/2632/files/858338/download?download_frd=1)
- 11.11 Strategy for Testing Series.pdf (https://imsa.instructure.com/courses/2632/files/858339?wrap=1) (https://imsa.instructure.com/courses/2632/files/858339/download?download_frd=1)

Unit 12 - Taylor Series

- 12.1 Power Series.pdf (https://imsa.instructure.com/courses/2632/files/858415?wrap=1) (https://imsa.instructure.com/courses/2632/files/858415/download?download_frd=1)
- 12.2 Proof of Interval of Convergence, Term-by -term Differentiation and Integration (Solution).pdf (https://imsa.instructure.com/courses/2632/files/858416?wrap=1) (https://imsa.instructure.com/courses/2632/files/858416/download?download_frd=1) *Revise
- 12.3 Representations of Functions as Power Series.pdf (https://imsa.instructure.com/courses/2632/files/858417?wrap=1) (https://imsa.instructure.com/courses/2632/files/858417/download?download_frd=1)
- 12.4 Taylor and Maclaurin Series.pdf (https://imsa.instructure.com/courses/2632/files/858419?wrap=1) (https://imsa.instructure.com/courses/2632/files/858419/download?download_frd=1)
- 12.5 New Maclaurin Series from Old.pdf (https://imsa.instructure.com/courses/2632/files/858422?wrap=1) (https://imsa.instructure.com/courses/2632/files/858422/download?download_frd=1)
- 12.6 Analytic Functions I.pdf (https://imsa.instructure.com/courses/2632/files/858420?wrap=1) (https://imsa.instructure.com/courses/2632/files/858420/download?download_frd=1)
- 12.7 Analytic Functions II.pdf (https://imsa.instructure.com/courses/2632/files/858412?wrap=1) (https://imsa.instructure.com/courses/2632/files/858412/download?download_frd=1)
- 12.8 Applications of Taylor Polynomials.pdf (https://imsa.instructure.com/courses/2632/files/858418?wrap=1) (https://imsa.instructure.com/courses/2632/files/858418/download?download_frd=1)
- Taylor Series Review.pdf (https://imsa.instructure.com/courses/2632/files/921804?wrap=1) (https://imsa.instructure.com/courses/2632/files/921804/download?download_frd=1) (Taylor Series Review SOLUTION.pdf (https://imsa.instructure.com/courses/2632/files/922216?wrap=1) (https://imsa.instructure.com/courses/2632/files/922216/download?download_frd=1))

Unit 13 - Polar Curves

- 13.1 Parametric Curves.pdf (https://imsa.instructure.com/courses/2632/files/858347?wrap=1) (https://imsa.instructure.com/courses/2632/files/858347/download?download_frd=1)
- 13.2 Tangents to Polar Curves.pdf (https://imsa.instructure.com/courses/2632/files/858423?wrap=1) (https://imsa.instructure.com/courses/2632/files/858423/download?download_frd=1)
- 13.3 Area and Arc Length in Polar Coordinates.pdf (https://imsa.instructure.com/

courses/2632/files/858424?wrap=1) (https://imsa.instructure.com/courses/2632/files/858424/download?download_frd=1)

Unit 14 - Vectors

- 14.1 Vector Basics.pdf (https://imsa.instructure.com/courses/2632/files/858425?wrap=1) (https://imsa.instructure.com/courses/2632/files/858425/download?download_frd=1)
- 14.2 Differentiation and Integration of Vectors.pdf (https://imsa.instructure.com/courses/2632/files/858343?wrap=1) (https://imsa.instructure.com/courses/2632/files/858343/download?download_frd=1)
- 14.3 Projectiles.pdf (https://imsa.instructure.com/courses/2632/files/858355?wrap=1) (https://imsa.instructure.com/courses/2632/files/858355/download?download_frd=1)
- 14.4 Vectors Review.pdf (https://imsa.instructure.com/courses/2632/files/858381?wrap=1) (https://imsa.instructure.com/courses/2632/files/858381/download?download_frd=1)

Problems of the Week

Semester 1

- POTW #1: Sup and inf (https://imsa.instructure.com/courses/2632/files/858511?wrap=1) (https://imsa.instructure.com/courses/2632/files/858511/download?download_frd=1)
- POTW #2: Bisection Method (https://imsa.instructure.com/courses/2632/files/858499?wrap=1) (https://imsa.instructure.com/courses/2632/files/858499/download?download_frd=1)
- POTW #3: Proving the Limit Laws (https://imsa.instructure.com/courses/2632/files/858478?wrap=1) (https://imsa.instructure.com/courses/2632/files/858478/download?download_frd=1)
- POTW #4: Continuity (https://imsa.instructure.com/courses/2632/files/858486?wrap=1) (https://imsa.instructure.com/courses/2632/files/858486/download?download_frd=1)
- POTW #5: Definition of the Derivative (https://imsa.instructure.com/courses/2632/files/858438?wrap=1) (https://imsa.instructure.com/courses/2632/files/858438/download?download_frd=1)
- POTW #6: Derivative as a Function (https://imsa.instructure.com/courses/2632/files/858481?wrap=1) (https://imsa.instructure.com/courses/2632/files/858481/download?download_frd=1)
- POTW #7: Related Rates (https://imsa.instructure.com/courses/2632/files/858487?wrap=1) (https://imsa.instructure.com/courses/2632/files/858487/download?download_frd=1)
- POTW #8: Maximize Angle (https://imsa.instructure.com/courses/2632/files/858484?wrap=1) (https://imsa.instructure.com/courses/2632/files/858484/download?download_frd=1)
- POTW #9: Proof of the EVT (https://imsa.instructure.com/courses/2632/files/858513?wrap=1) (https://imsa.instructure.com/courses/2632/files/858513/download?download_frd=1)
- POTW #10: Global Maximum (https://imsa.instructure.com/courses/2632/files/858498?wrap=1) (https://imsa.instructure.com/courses/2632/files/858498/download?download_frd=1)
- POTW #11: Construction of the Definite Integral, Part I (https://imsa.instructure.com/courses/2632/files/858509?wrap=1) (https://imsa.instructure.com/courses/2632/files/858509/

download?download_frd=1)

- POTW #12: Construction of the Definite Integral, Part 2 (https://imsa.instructure.com/courses/2632/files/858510?wrap=1) (https://imsa.instructure.com/courses/2632/files/858510/download?download_frd=1)

Semester 2

- POTW #13: Darboux Integrable implies Riemann Integrable (https://imsa.instructure.com/courses/2632/files/887893?wrap=1) (https://imsa.instructure.com/courses/2632/files/887893/download?download_frd=1)
- POTW #14: Proofs of Error Bounds (https://imsa.instructure.com/courses/2632/files/858311?wrap=1) (https://imsa.instructure.com/courses/2632/files/858311/download?download_frd=1)
- POTW #15: Volume of Solids of Revolution and Arc Length of Curves (https://imsa.instructure.com/courses/2632/files/898490?wrap=1) (https://imsa.instructure.com/courses/2632/files/898490/download?download_frd=1)
- POTW #16: Proofs of Convergence Tests for Improper Integrals (https://imsa.instructure.com/courses/2632/files/901679?wrap=1) (https://imsa.instructure.com/courses/2632/files/901679/download?download_frd=1)
- POTW #17: Proof of the Integral Test (See worksheet 11.5)
- POTW #18: Prove Taylor's Theorem (See worksheet 12.7)

Errata for the handouts - **Please alert me to any typos you find by entering them in the document below.**

- Typos in BC Handouts (https://docs.google.com/document/d/1G86XVKfhD993ByqNdwAEn6_0Rbi0A-HCoi7QGS_30kA/edit?usp=sharing)

Old stuff from semester 1

Theory Portfolio - **We are not doing a portfolio in BC 2/3**

- template.tex (https://imsa.instructure.com/courses/2632/files/858458?wrap=1) (https://imsa.instructure.com/courses/2632/files/858458/download?download_frd=1) (upload to overleaf)
- BC I_II Portfolio Specifications.pdf (https://imsa.instructure.com/courses/2632/files/858516?wrap=1) (https://imsa.instructure.com/courses/2632/files/858516/download?download_frd=1)
- Su_Some Guidelines for Good Mathematical Writing.pdf (https://imsa.instructure.com/courses/2632/files/858464?wrap=1) (https://imsa.instructure.com/courses/2632/files/858464/download?download_frd=1)

BC I_II S24 Final Exam Information and Checklists.pdf (https://imsa.instructure.com/courses/2632/files/858512?wrap=1) (https://imsa.instructure.com/courses/2632/files/858512/download?download_frd=1)

BC I/II Syllabus (https://imsa.instructure.com/courses/2632/files/858469?wrap=1) (https://

imsa.instructure.com/courses/2632/files/858469/download?download_frd=1)

BC I/II Learning Standards / Grade Record (https://imsa.instructure.com/courses/2632/files/858494?wrap=1) (https://imsa.instructure.com/courses/2632/files/858494/download?download_frd=1)